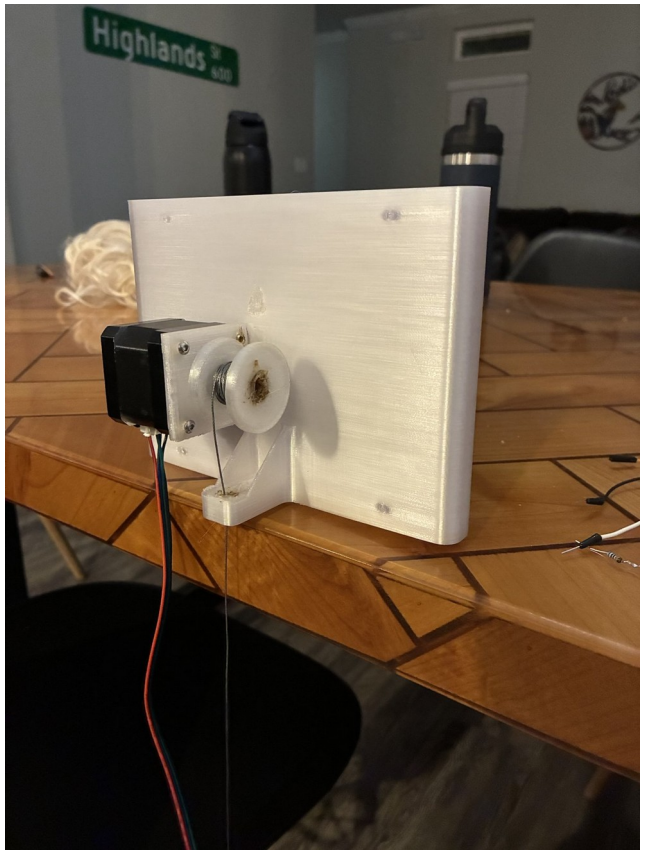
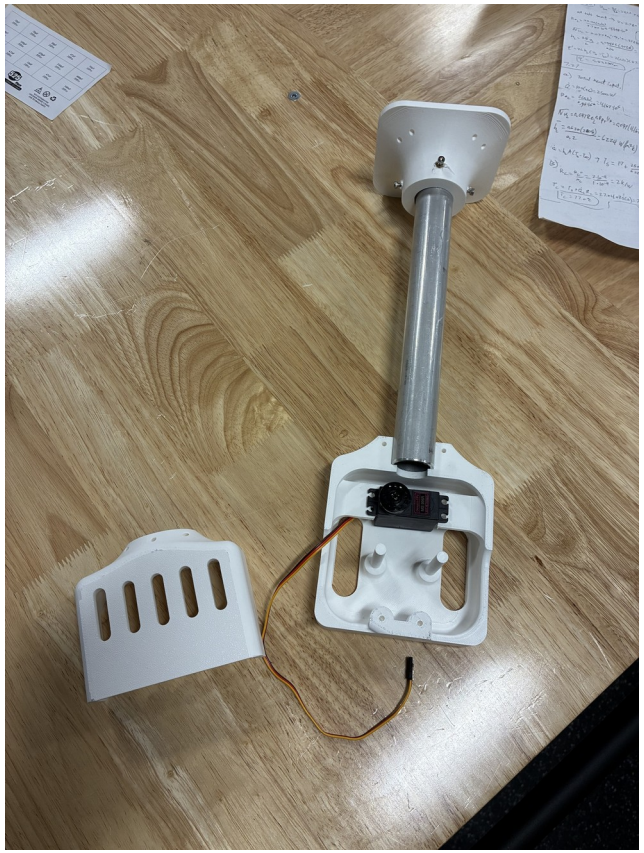
RoomCleaner

A cable robot that picks laundry off the floor

Jasper Buntinx
Mechanical Engineering, Texas A&M University
October 27, 2026





About Me

- **Education:** Mechanical Engineering, Texas A&M · *[graduation: Month 20XX]*
- **EcoFil:** filament recycler (shredder, dryer, extruder)
- **Sculpture:** Austin Creek Show · fiber laser, welding
- **Outside class:** *[hobbies, interests]*
- **Why this project:** *[one line]*

[Photo of you building: EcoFil, a sculpture, or the workshop]

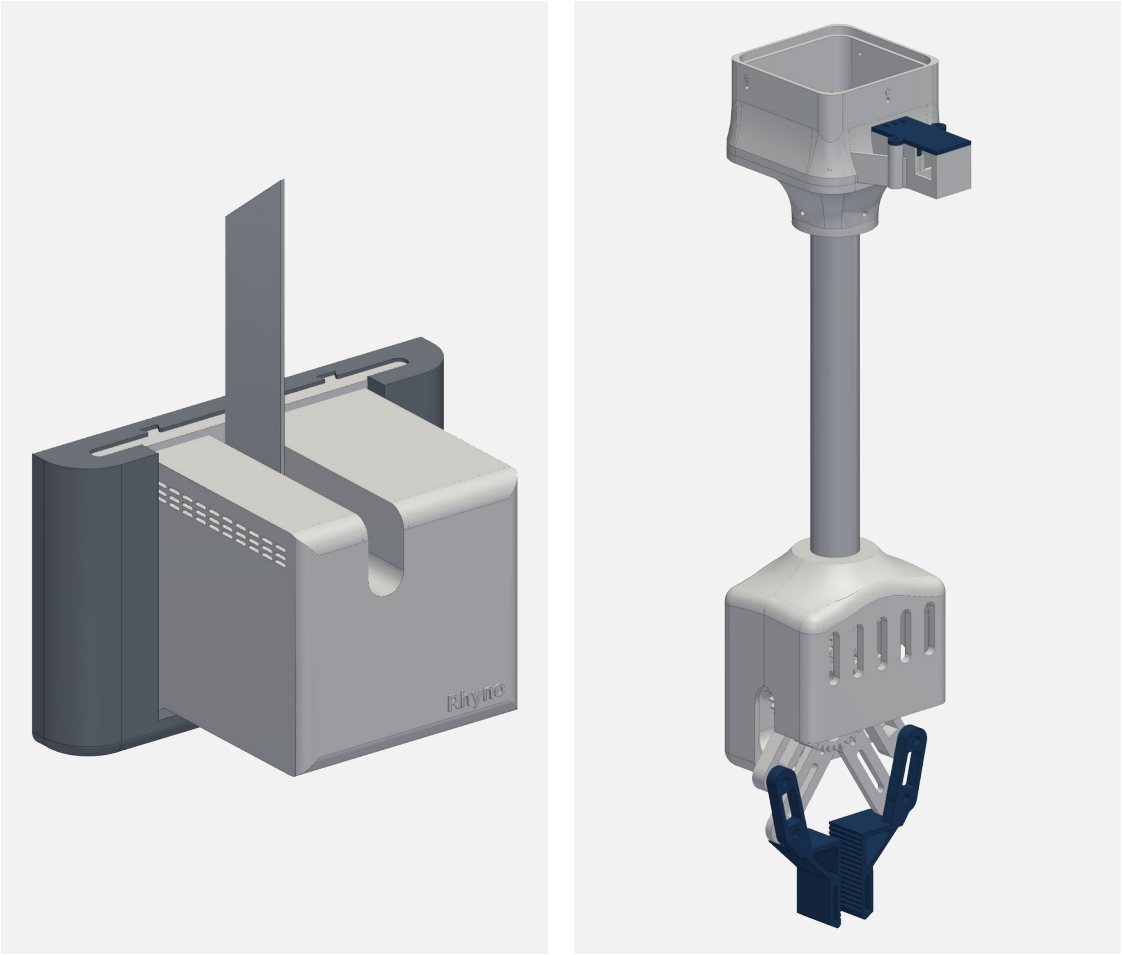
The Goal

- **Task:** floor → hamper, 11 × 9 ft room
- **How:** 4 ceiling winches, 1 claw on cables
- **Load:** jeans 0.9 kg at 2 m/s² · test 1 lb
- **Build:** printed PETG (polyethylene terephthalate glycol), hobby budget

What “done” means

Gate 1	Gate 2	Gate 3
One station runs repeatably	Stations hold and move 1 lb	One garment into the hamper

Now: gate 1 passed on the bench, gate 2 is this week’s test



Final design (CAD): a wall station and the claw

Scope and Timeline

Jul 16

Concept
Requirements, simulator, parts list

Design

Aug 6 – Sep 2

Gen 1
Corner plates, one central controller, finger claw

Design · Build · Test

Sep – Oct 6

Rethink
Sensors per winch, wiring, line outlets

Design

Oct 7 – 21

Gen 2 → final
SolidWorks: station v1→v3, spool v1→v3, claw → four-bar

Design · Build · Test

Oct 22 – 27

Build and test
Print, wire, three-station test

Build · Test

Me: a solo project (team of one)

- Requirements, calculations and test plans
- All CAD (computer-aided design) in SolidWorks: stations, spool, claw, fin
- 3D printing and fits; the heat fin on the shear and manual mill
- Wiring, soldering, electronics bring-up and debugging
- Control software: *[confirm the parts you wrote]*

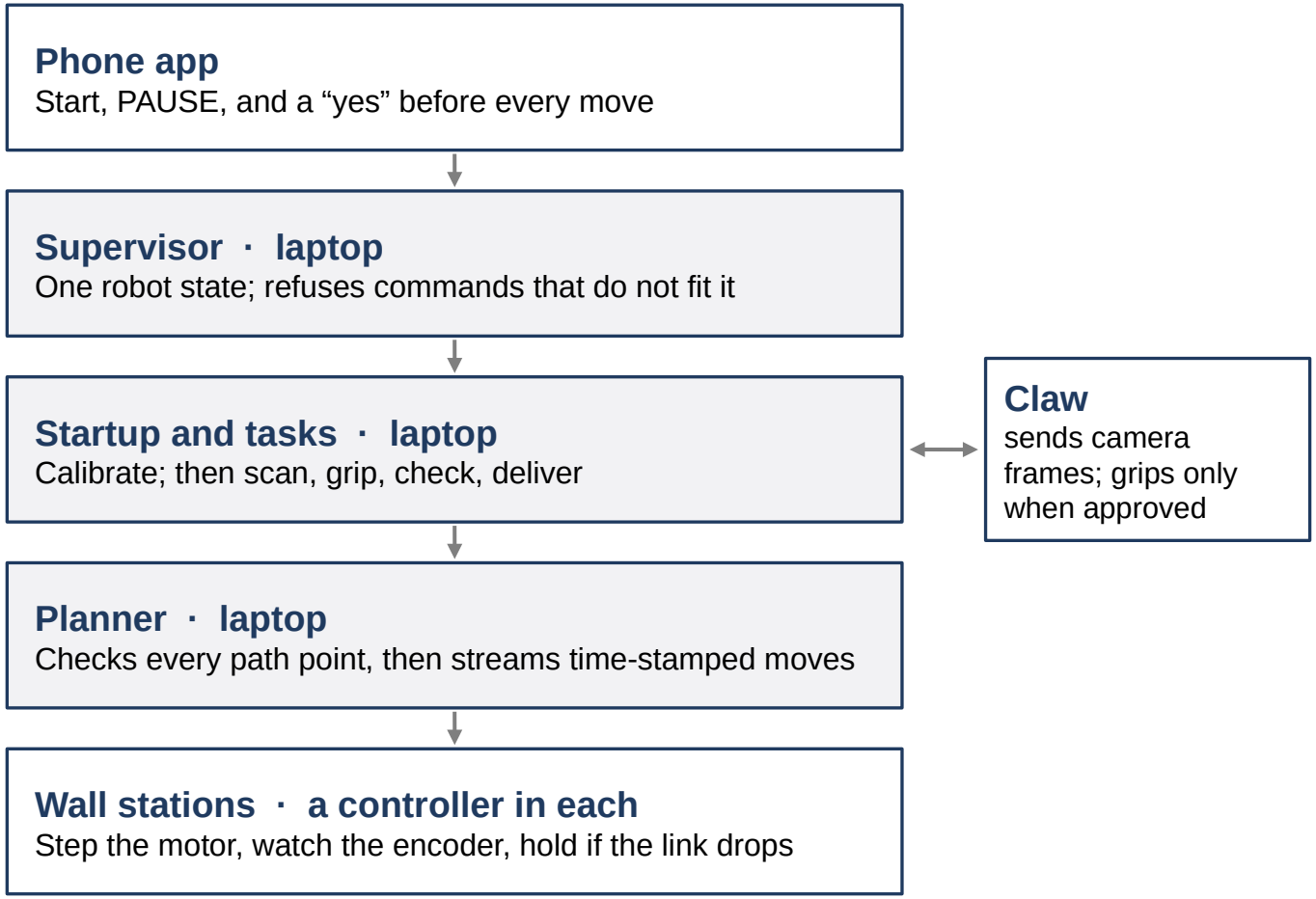
Bought or outside

- Stepper motors, drivers, controller boards
- Servo, camera, battery, encoder boards
- Garment detector: YOLO-World, an open-source model
- Help: *[shop access or mentors, if any]*

How It Works

Simulated end to end

- **4 wall stations:** winch, encoder, fin, controller
- **1 claw:** gripper, camera, battery
- **Laptop:** plans + checks every move
- **Phone:** start · PAUSE · approve



Commands flow down; status and camera frames flow back up.

What I’m Optimizing

NOT MET

Lift capacity

Maximize

Target: Claw + jeans: 1.76 kg

Now: 1.56 kg (calculated)

NOT MET

Claw mass

Minimize

Target: 0.45 kg budget

Now: ≈ 0.86 kg (CAD) · *[weighed]*

TIGHT

Motor temperature

Limit

Target: Under 60 °C (PETG)

Now: 58 °C prototype · 45–49 °C 1/8 in fin

MET IN DESIGN

Working area

Maximize

Target: Whole 11 × 9 ft floor

Now: All but 0.2 m at walls (simulated)

MET IN DESIGN

Wiring

Minimize

Target: One plug per station

Now: One 12 V pair · 0.51 V drop

TEST PENDING

Accuracy

Maximize

Target: ≤ 2 mm per cable

Now: 0.7 mm floor board (calculated)

TEST PENDING

Speed

Maximize

Target: 50 mm/s · 2 m/s²

Now: Designed for; not yet run

TEST PENDING

Cost

Minimize

Target: Hobby budget

Now: *[total: $___]*

Not traded: safety. Anchors in studs, one hard stop, holds on any fault, nobody under the robot

Trade-offs

Motor current ↑

Gain: pull, T ∝ I (+41 % at 1.5 A)

Cost: heat, P = I²R (2.0 → 4.1 W)

Set: 1.5 A, vented shell + fin

Claw strength ↑

Gain: gear safety factor 0.4 → 2

Cost: mass; payload = capacity − claw

Set: module 1.5 gears; mass budget open

Room size ↑

Gain: more floor covered

Cost: longer wire, more resistance: R = ρL/A

Set: only 12 V travels, on 16-gauge wire

Reach toward the ceiling ↑

Gain: more working volume

Cost: tension, T = W / (4 sin φ): 12 → 32 N

Set: stop 0.4 m below the anchors

Spool radius ↓

Gain: pull, T = τ / r

Cost: line speed, v = ω r; less line per layer

Set: Ø15 core: 34 N empty, 25 N full

Safety margins ↑

Gain: no skipped steps, no wall contact

Cost: usable force (÷ 1.5) and floor area

Set: ÷ 1.5; 0.2 m walls, 0.4 m ceiling

How they chain:

Claw mass ↑ → **Pull needed ↑** → **Motor current ↑** → **Heat ↑ as I²** → **Fin and vents**

Cable Tension

Calculated

- **Cables only pull:** $T > 0$
- Tension climbs near the ceiling
- 32 N at 0.15 m · 8.3 N at 0.6 m
- **Decision:** stay 0.4 m below anchors

Equation

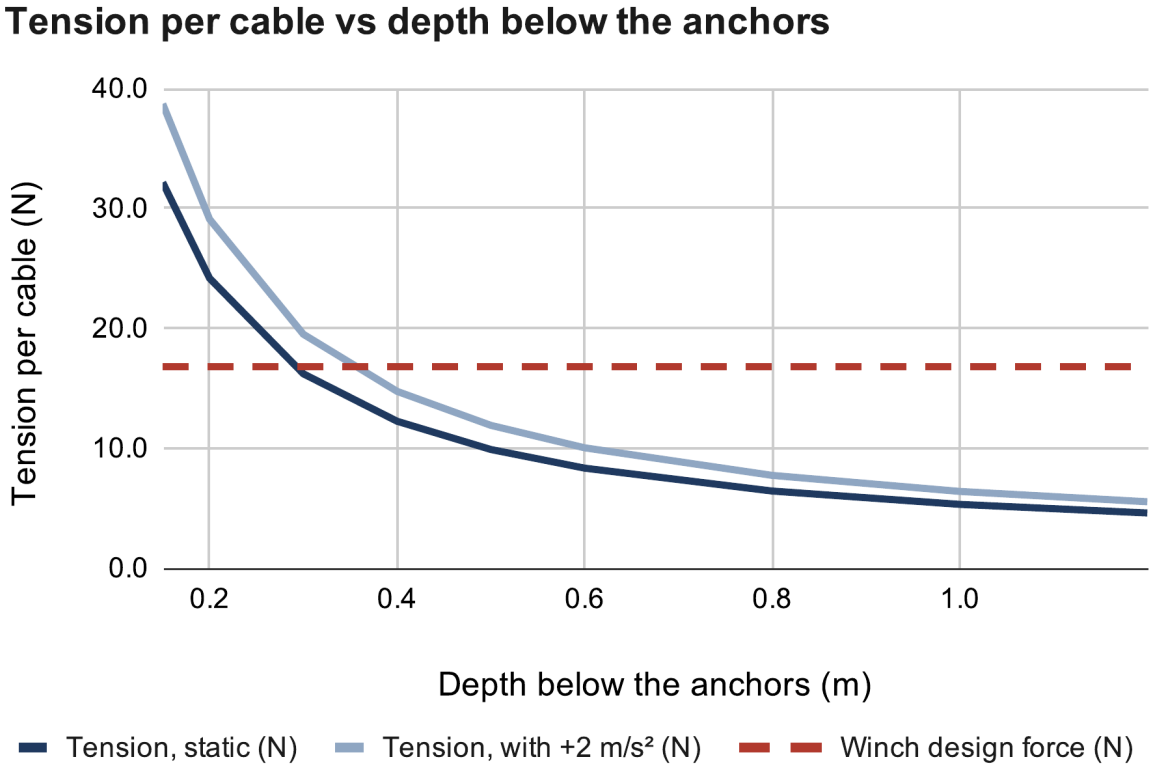
$$T = W / (4 \sin\varphi), \quad \sin\varphi = d / \sqrt{(d^2 + D_h{}^2)}$$

$$\sin\varphi = 0.4 / \sqrt{(0.4^2 + 2.17^2)} = 0.181$$

$$T = 8.87\ \text{N} / (4 \times 0.181) = \mathbf{12.2\ N}$$

$$\times (1 + a/g) = 12.2 \times 1.20 = \mathbf{14.7\ N} < 16.8\ \text{N}$$

where W = (0.45 + 0.454) kg × 9.81 m/s² = 8.87 N · d = 0.4 m below the anchors · D_h = 2.17 m centre to corner · a = 2 m/s² · 16.8 N = winch design force



My calculation sheet (Google Sheets), “Cable tension” tab. Calculated, not measured.

Winch Pull

1.5 A set · pull calculated

- NEMA 17 stepper (1.7 in frame)
- Current: 1.06 → **1.5 A**
- Pull: 34 N empty · 25 N full spool
- **Next:** luggage-scale pull test

Equation

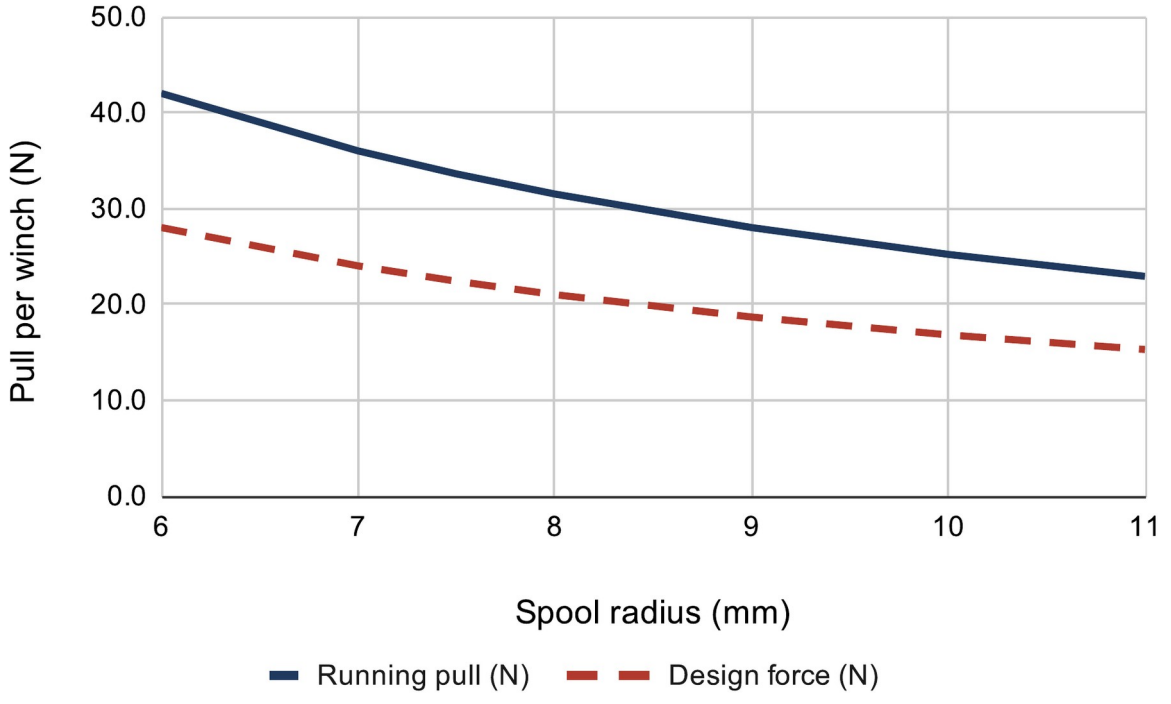
$$I = V_{ref} / (8\,R_s) = 1.2\ V / (8 \times 0.1\ \Omega) = \mathbf{1.5\ A}$$

$$\tau_{run} = 0.7 \times 0.36\ N{\cdot}m \times 1.5 / 1.5 = 0.252\ N{\cdot}m$$

$$T = \tau / r = 0.252 / 0.010\ m = \mathbf{25.2\ N}$$

$$F_{design} = 25.2\ N / 1.5 = \mathbf{16.8\ N}$$

where V_{ref} = driver reference voltage · R_s = 0.1 Ω sense resistor · 0.36 N·m holding torque at 1.5 A rated · 0.7 = running / holding (assumed) · r = 10 mm full spool · 1.5 = skip margin



My calculation sheet (Google Sheets), “Winch pull” tab. Calculated, not measured.

One Controller per Winch

Built

- **Gen 1:** one board, four motors
- Sensors per winch: 10 wires per corner
- **Decision:** controller in every station
- **Result:** 5× less wire

Equation

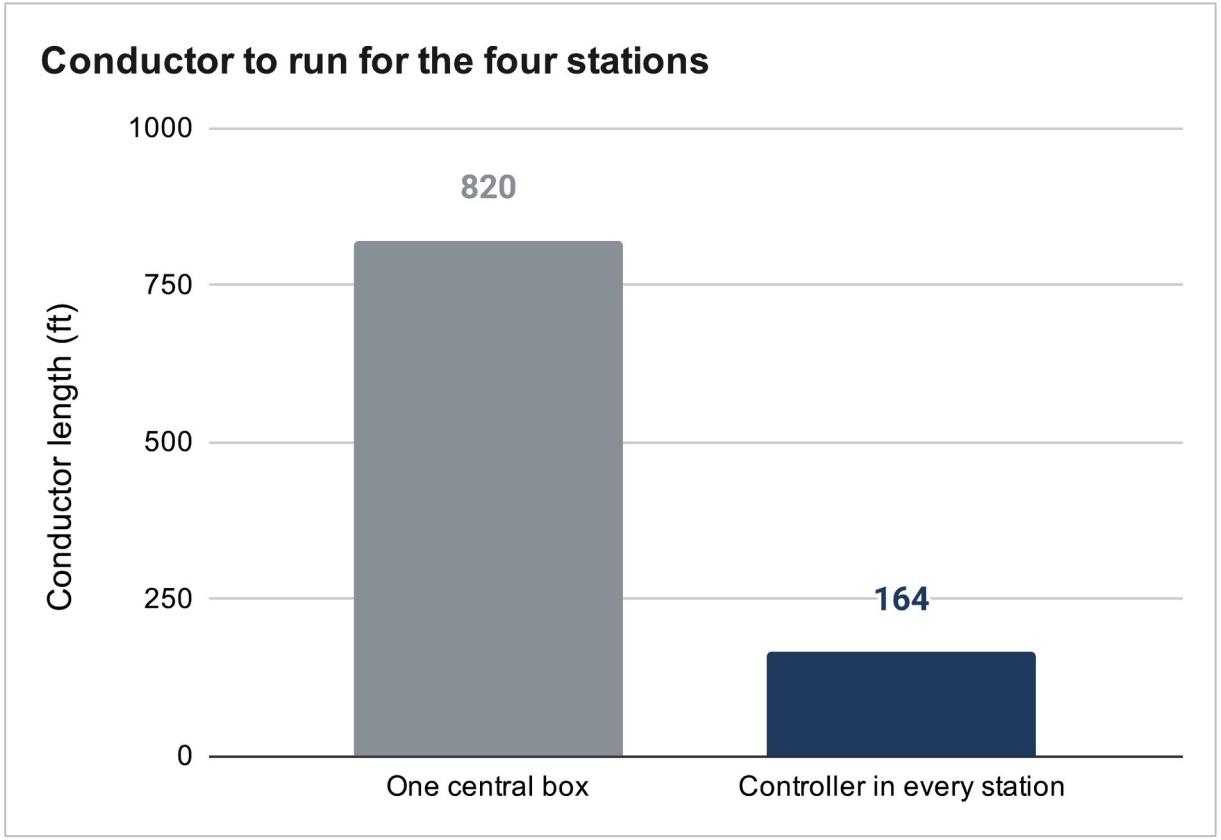
Wire = n × ΣL

Central box: 10 × 82 ft = **820 ft**

Controller per station: 2 × 82 ft = **164 ft**

Sync error: $\Delta x = v\,\Delta t = 50\ \text{mm/s} \times 50\ \text{ms} =$ **2.5 mm**

where n = conductors per corner (motor 4, switch 2, encoder 4) · ΣL = 6 + 21 + 23 + 32 ft of routes · v = line speed · Δt = clock skew between stations



My calculation sheet (Google Sheets), “Wiring” tab. Calculated, not measured.

Cooling the Motor

Fin built · temperature calculated

- Sealed PETG shell traps motor heat
- **Limit:** PETG creeps above 60 °C
- **Decision:** aluminium fin through the shell
- **Next:** 90-min hold test at 1.5 A

Equation

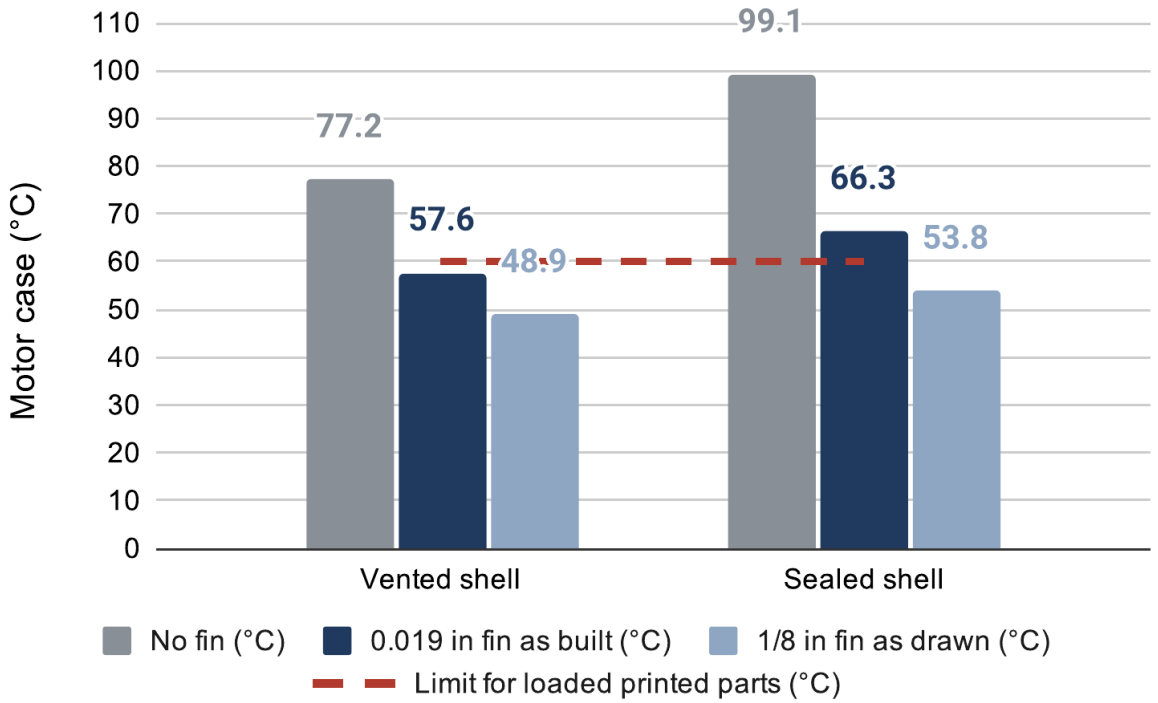
$P = I^2 R_{coil} = (1.5\ A)^2 \times 1.8\ \Omega = \mathbf{4.05\ W}$

$1/R_{th} = 1/R_{shell} + 1/R_{fin}, \quad \Delta T = P \times R_{th}$

0.019 in fin: $R_{fin} = 23.1\ K/W \rightarrow \mathbf{57.6\ °C}$

1/8 in fin: $R_{fin} = 11.9\ K/W \rightarrow \mathbf{48.9\ °C}$

where R_{coil} = 1.8 Ω per phase (assumed) · R_{shell} = 12 + 1 K/W (motor → shell air → room) · room 23 °C · 1.5 W of electronics in the shell · limit 60 °C



My calculation sheet (Google Sheets), “Motor temperature” tab. Calculated, not measured.

Heat Fin Sizing

Calculated

- **Prototype 0.019 in:** done by 136 mm
- **Design 1/8 in:** 74 % efficient at 200 mm
- **Built length:** 200 mm

Equation

$m = \sqrt{(2h / (k\,t))}$, $L_{95} = 1.83 / m$

0.019 in: $m = \sqrt{(2 \times 7 / (160 \times 0.00048))} = 13.5\ /m$

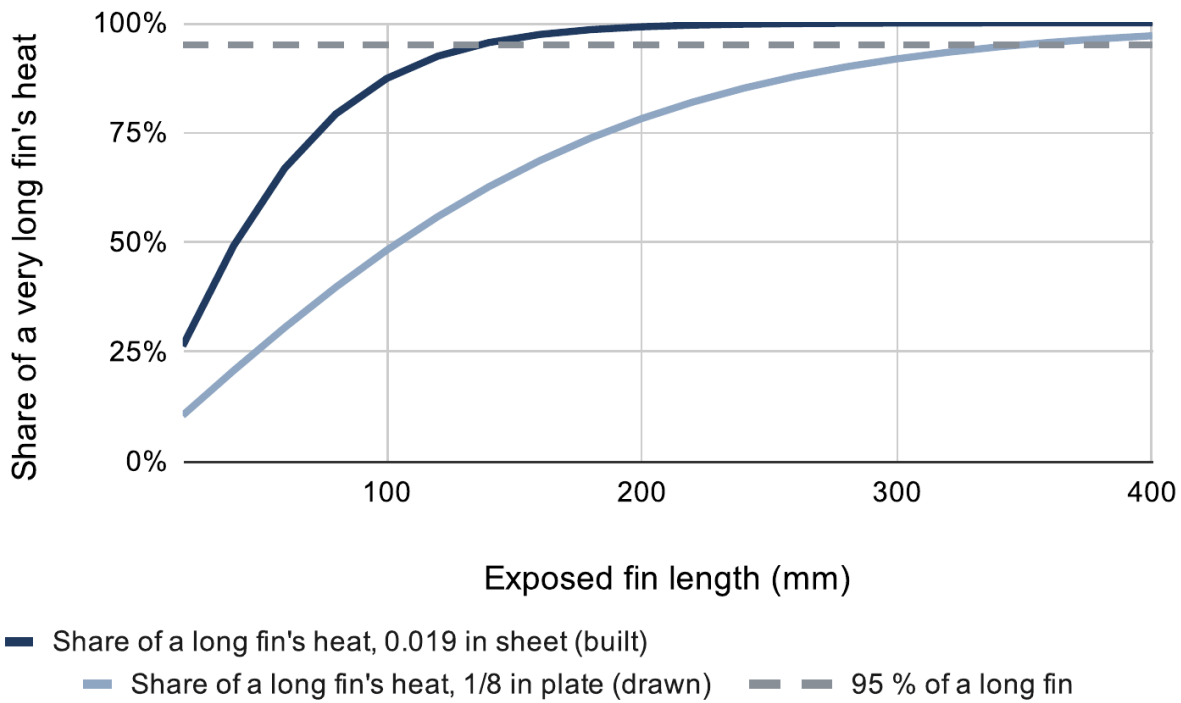
$\rightarrow L_{95} = 1.83 / 13.5 =$ **136 mm**

1/8 in: $m = 5.25\ /m \rightarrow L_{95} =$ **349 mm**

$\eta = \tanh(mL) / (mL) = \tanh(1.05) / 1.05 =$ **0.74**

$R_{fin} = 1 / (7 \times 0.74 \times 0.0176) =$ **10.9 K/W**

where $R_{fin} = 1 / (h\,\eta\,A)$ · h = 7 W/m²·K still air · k = 160 W/m·K aluminium · t = thickness · L = 200 mm · A = 2 × 44 mm × 200 mm = 0.0176 m² · L_{95} = 95 % of a long fin's heat



My calculation sheet (Google Sheets), “Heat fin” tab. Calculated, not measured.

Homing

Calculated · simulated

- **Plan A:** bead trips a switch at the outlet
- Cables can't get shorter than at home
- **Decision:** camera + floor board calibration
- **Result:** whole room · beads = end stops

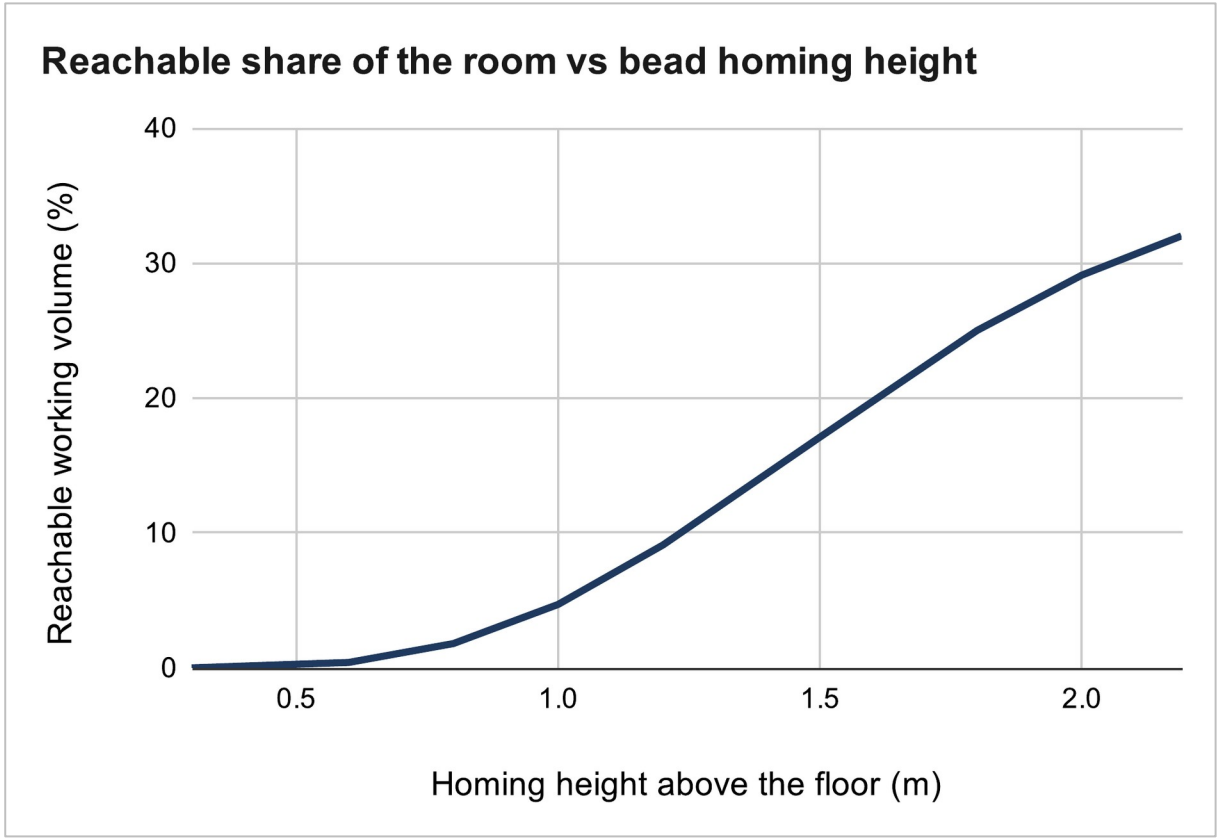
Equation

$$L_i(\mathbf{p}) \geq L_{i,\text{home}} = \|\mathbf{a}_i - \mathbf{p}_{\text{home}}\|$$

Home 1 m up (room centre) → **5 %** reachable

Home 2 m up → **29 %** reachable

where a_i = anchor i · p = claw position · reachable = every cable at or above its home length, inside 0.2 m wall and 0.4 m ceiling margins



My calculation sheet (Google Sheets), “Bead homing” tab. Calculated, not measured.

The Claw

Body built · hands in CAD

- **Gen 1:** 5 TPU (thermoplastic polyurethane) fingers
- Couldn't pinch fabric · 4.6× over mass
- **Final:** geared four-bar, 53 mm opening

Equation

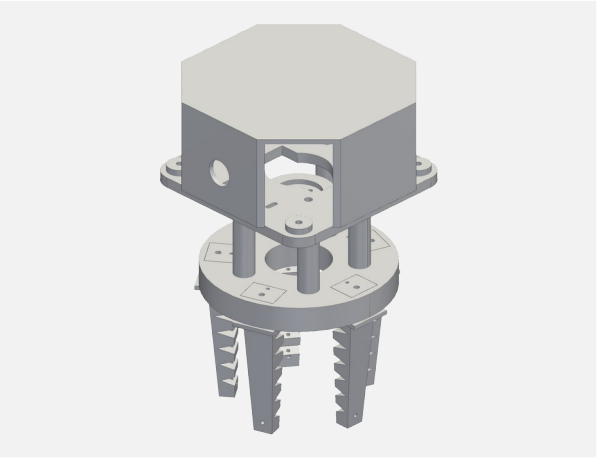
$$N \geq m\,g\,/\,2\mu = 0.454 \times 9.81\,/\,0.6 = \mathbf{7.4\ N}$$

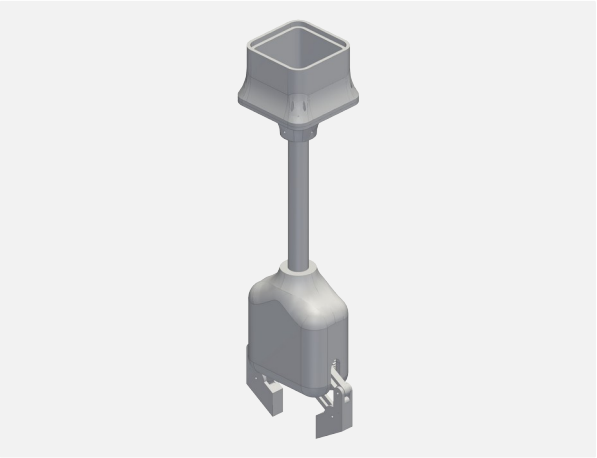
$$F_t = 2\tau\,/\,d = 2 \times 1.078\,/\,0.027 = 79.9\ N$$

$$\sigma = F_t\,/\,(b\,m\,Y) = 79.9\,/\,(15.7 \times 1.5 \times 0.309)$$

$$\sigma = 11.0\ MPa \rightarrow SF = 22.5\,/\,11.0 = \mathbf{2.05}$$

where N = grip per jaw · μ = 0.3 · τ = 1.078 N·m stall · d = 27 mm · b = 15.7 mm · m = 1.5 mm · Y = 0.309 · allowable 22.5 MPa · module 1: SF 0.4









Fingers → v6 clamp → final geared four-bar (CAD) → printed gear train with the servo

Weight: the Open Problem

Open problem

- **Capacity at 1.5 A:** 1.56 kg
- **Claw:** ≈ 0.86 kg · *[weighed: ___ g]*
- 1 lb test fits · jeans 0.2 kg over
- **Fix:** lighter claw or 3:1 winch

Equation

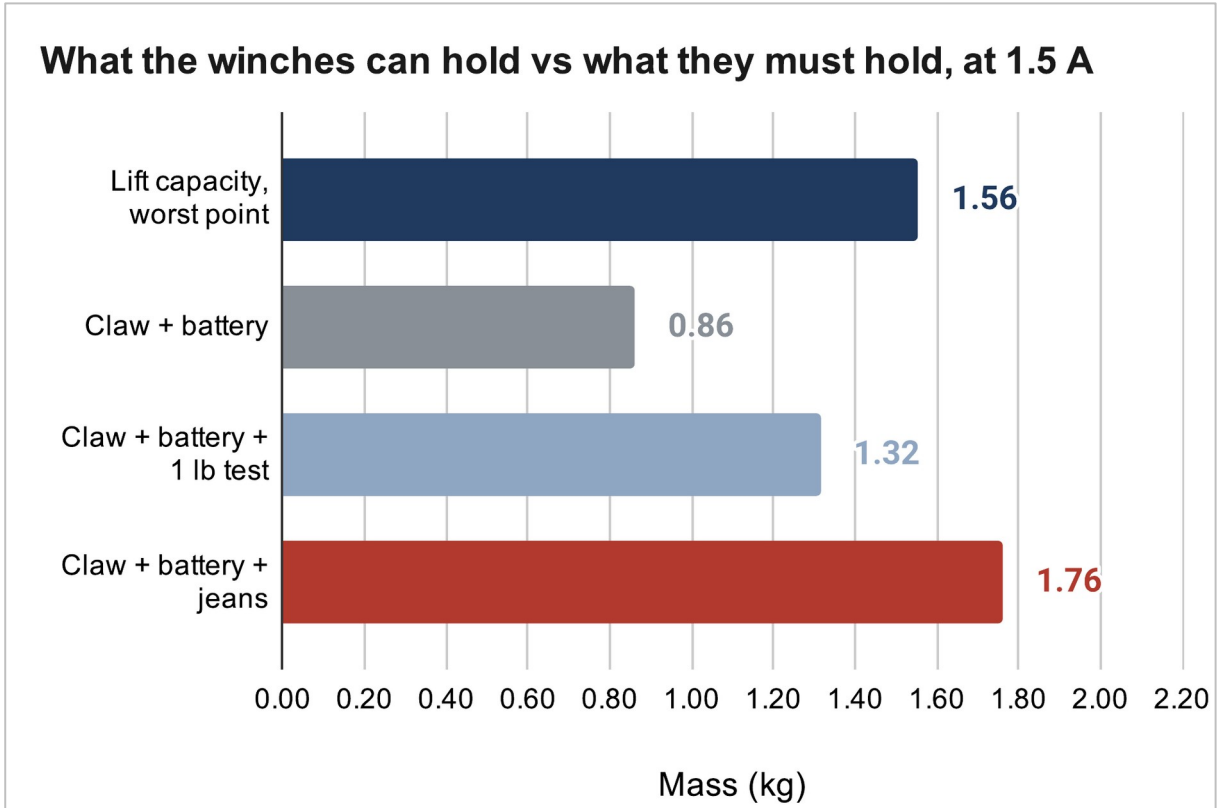
$$m_{max} = F_{design} / (k\,g)$$

$$= 16.8\ N / (1.101 \times 9.81\ m/s^2) = \mathbf{1.56\ kg}$$

$$m_{claw} = 0.55 \times 0.927 + 0.198 + 0.055 = \mathbf{0.76\ kg}$$

$$\text{Need: } 0.76 + 0.10 + 0.90 = \mathbf{1.76\ kg} > 1.56\ kg$$

where k = worst cable tension per unit weight (+2 m/s²), linear program over the room · 0.55 = print fill · 0.927 kg solid PETG (CAD) · tube 0.198 · servo 0.055 · battery 0.10 · jeans 0.90 kg



My calculation sheet (Google Sheets), “Lift budget” tab. Calculated, not measured.

Printed Fits

Built

- PETG · Creality K1 · 0.2 mm layers
- Printed holes come out small
- **Spool:** 3 prints to fit the shaft
- **Lesson:** fit coupon first

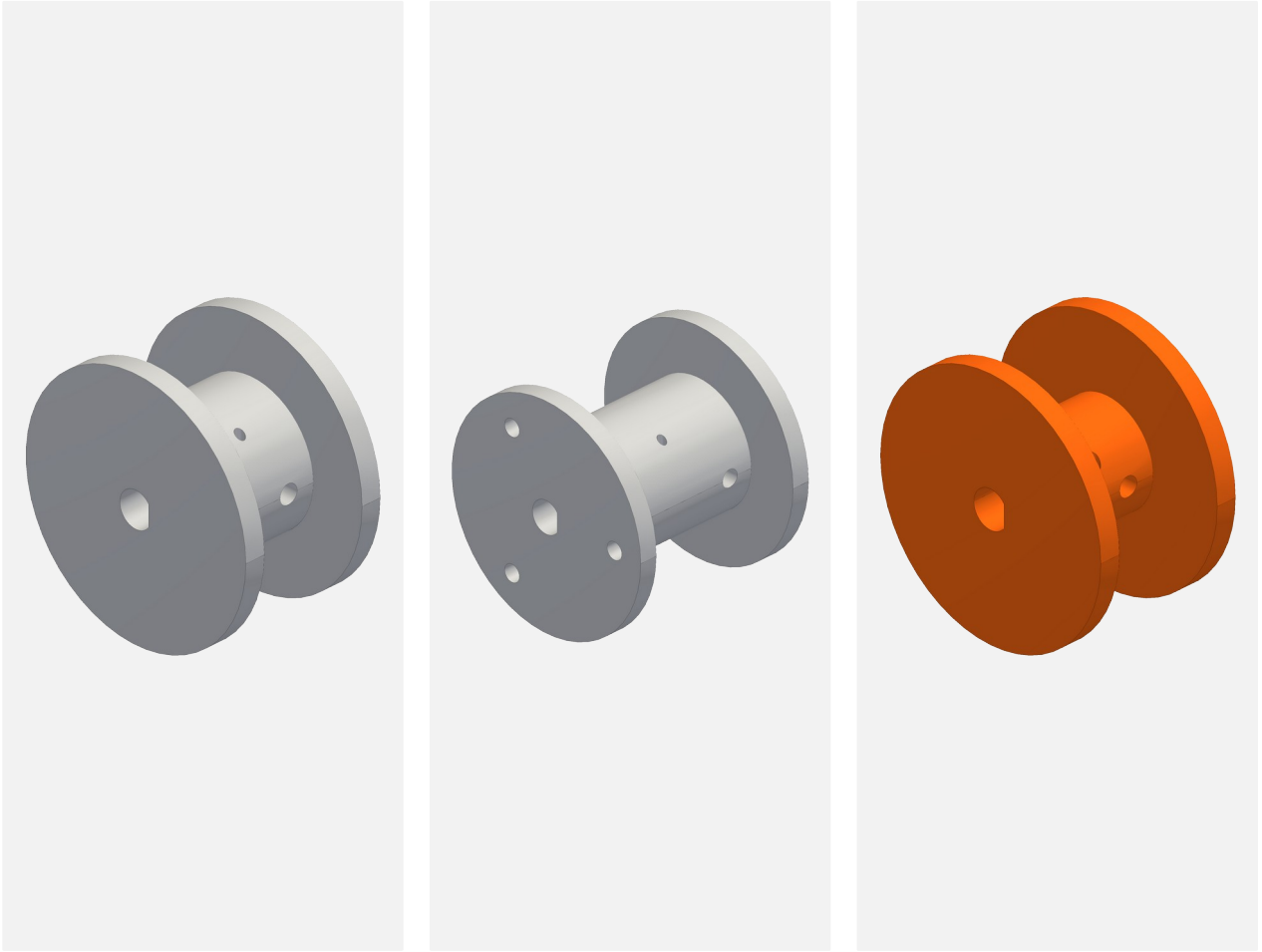
Equation

$$c = (D_{hole} - u) - D_{shaft}$$

$$Ø5.2 \text{ too tight} \rightarrow u \geq 0.2 \text{ mm}$$

$$c = (5.45 - 0.2) - 5.00 = \mathbf{0.25\ mm}$$

where c = clearance · u = printer undersize · D_{shaft} = 5.00 mm motor shaft · v3 bore Ø5.45 with a D-flat; the set screw on the flat carries the torque



Spool v1, v2, v3 (CAD)

Machining the Fin

Built

- **Shop drawing first:** 1:1 template
- **Prototype:** 0.019 in sheet on hand
- **One mill setup:** 5 holes, ±0.1 mm
- Bolted on, no reaming







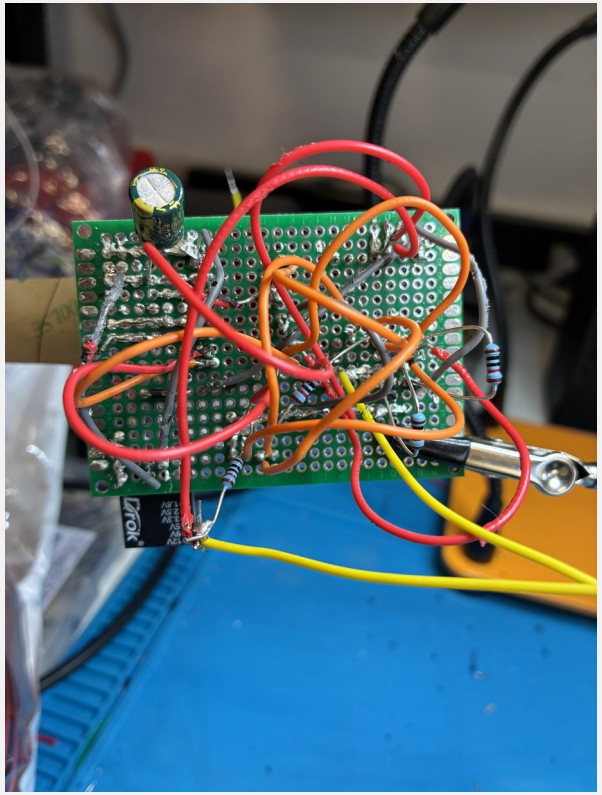


Shear and blanks · mill setup with the drawing · drilling · the finished fin

Electronics Bring-Up

Measured

- **Boots safe:** driver off until armed
- **Schematic = soldered board:** 16 nets
- **Failure:** logic rail 1.14 V, not 3.3 V
- **Now:** meter before code





Soldered station board (left); bench bring-up (right)

First Test

Planned · this week

- **3 stations:** fewest cables that hold a point
- Self-check → calibrate → move → pickup
- **Load:** 1 lb, then one garment
- Pass limits written first

[Photo or video from the three-station test]

Check	Pass limit	Result
Startup self-check	All stations answer; slack taken up	*[fill in]*
Calibration fit	Held-out poses inside the safety margin	*[fill in]*
Short move, 1 lb	300 mm at 50 mm/s, ≤ 2 mm per cable	*[fill in]*
Pickup	One garment into the hamper	*[fill in]*

Thermal Check

Calculated · simulation next

- **Limit:** PETG mount < 60 °C
- **Hand calc:** 56–59 °C (0.019 in) · 45–49 °C (1/8 in)
- **SolidWorks:** motor, fin, screws, mount
- **Hold test:** 90 min at 1.5 A

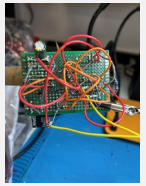
[SolidWorks temperature plot: motor, fin, screws and mount]

Method	0.019 in fin	1/8 in fin	PETG mount
Hand calc	56–59 °C	45–49 °C	≈ motor − 2 K
SolidWorks	*[fill in]*	*[fill in]*	*[fill in]*
Hold test	*[fill in]*	*[fill in]*	*[fill in]*

Three ways to know one temperature: calculate it, simulate it, measure it

Good and Poor Decisions

Good decisions



One controller per winch
5× less wire; each station tested alone



Slide-in wall mount
A station comes off; the anchor stays put



Camera calibration over bead homing
Whole room reachable; no tape measure

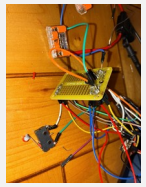


Gear strength checked at stall
Module 1 would have failed; fixed in CAD

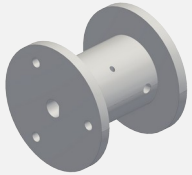
Poor decisions



Mass budget set late
Claw + jeans is 0.2 kg over, even at 1.5 A



Bead homing as plan A
Under a third of the room; a redesign



Spool printed before a fit coupon
Three prints to fit the shaft



Several stations before one was proven
Every fix repeated on each station

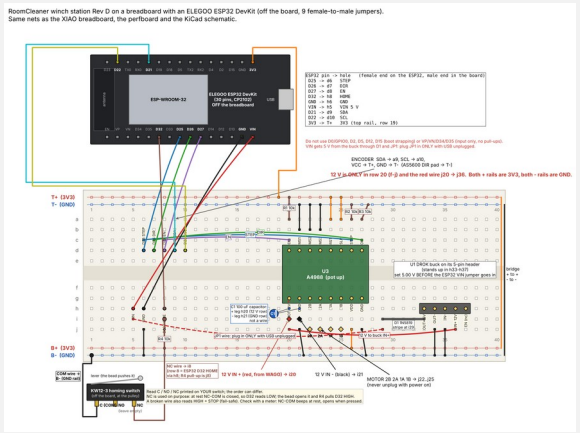
Thorough vs Scrappy

Thorough



- **Heat fin:** equations → shop drawing → one mill setup, ±0.1 mm
- **Station board:** schematic checked net by net against the solder
- **Gears:** strength checked at servo stall before printing

Scrappy



- **Fried a controller:** ESP32 on nine jumpers, same firmware
- **Floor board:** paper tags, 250 mm grid
- **Prototype fin:** 0.019 in sheet on hand; design is 1/8 in
- **Three stations** tested while the fourth is built

My rule: thorough where a mistake is expensive or unsafe; scrappy where it is cheap to undo

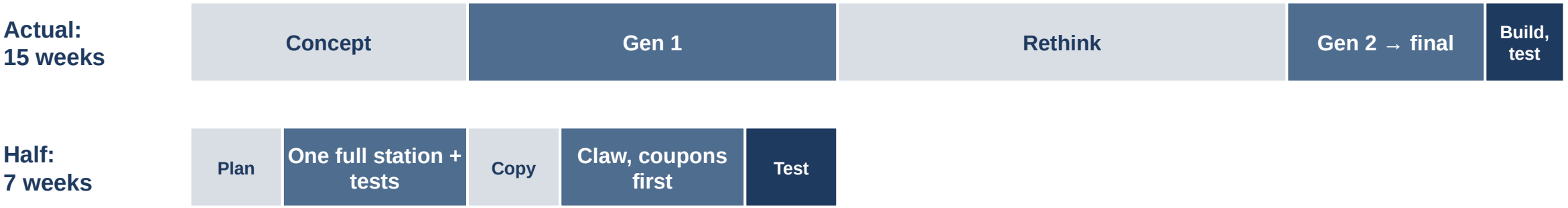
Lessons Learned

Do differently

- **Weigh early:** budget from winch pull
- **Coupons first:** fit before part
- **One station first:** prove, then copy
- **Meter before code**

In half the time

- **Freeze** requirements in week 1
- **Order** long-lead parts on day 1
- **Start** with 3 stations
- **One generation** per part, test-gated



Same scale. Actual: Jul 16 to Oct 27. The half-time plan proves one station before copying it.

Questions

Jasper Buntinx · jasper_buntinx@tamu.edu

Where It Stands

Part	Status	Next test
Wall stations	Built; bench-commissioned *[how many]*	Three-station test, then the fourth
Station electronics	Built; drivers at 1.5 A *[how: Vref reading]*	90-minute hold test at 1.5 A
Heat fin	Prototype 0.019 in built; 1/8 in designed (D38)	Machine the 1/8 in fin; hold test
Thermal model of the station	Hand calc at 1.5 A; SolidWorks model set up	Run it; then the hold test
Claw body: tube, mount, servo, gears	Built	Weigh it
Claw hands, camera housing, encoder casing	Checked in CAD; printing	Fit checks on the first print
Calibration and control software	Simulated end to end	First hardware run
Winch pull	Calculated at 1.5 A: 25–34 N	Luggage-scale pull test

Fin Sizing for 4 W

Calculated

- Fin alone carries all 4.05 W
- **0.019 in:** never (max ≈ 2 W)
- **1/8 in:** 145–185 mm
- Thickness beats length

Painted 6061, 44 mm wide	Mass	Motor, 200 mm, fin alone	Shortest for ≤ 60 °C
0.019 in (prototype)	14 g	100–120 °C	never (max ≈ 2.0 W)
1/16 in	47 g	66–72 °C	never (max ≈ 3.6 W)
0.090 in	68 g	60–65 °C	210 mm or never
1/8 in (design, D38)	94 g	56–59 °C	145–185 mm
3/16 in	141 g	53–55 °C	125–140 mm
1/4 in	189 g	51–52 °C	115–125 mm

All 4.05 W through the fin, 23 °C room, paste contact; ranges: flange at one temperature to heat spread over the patch. Calculated.

Equation

$$q_\infty = w\sqrt{(2\,h\,k\,t)} \times \theta_b$$

$$0.019\text{ in: } 0.044\sqrt{(2 \cdot 9.3 \cdot 167 \cdot 0.000483)} \times 37$$

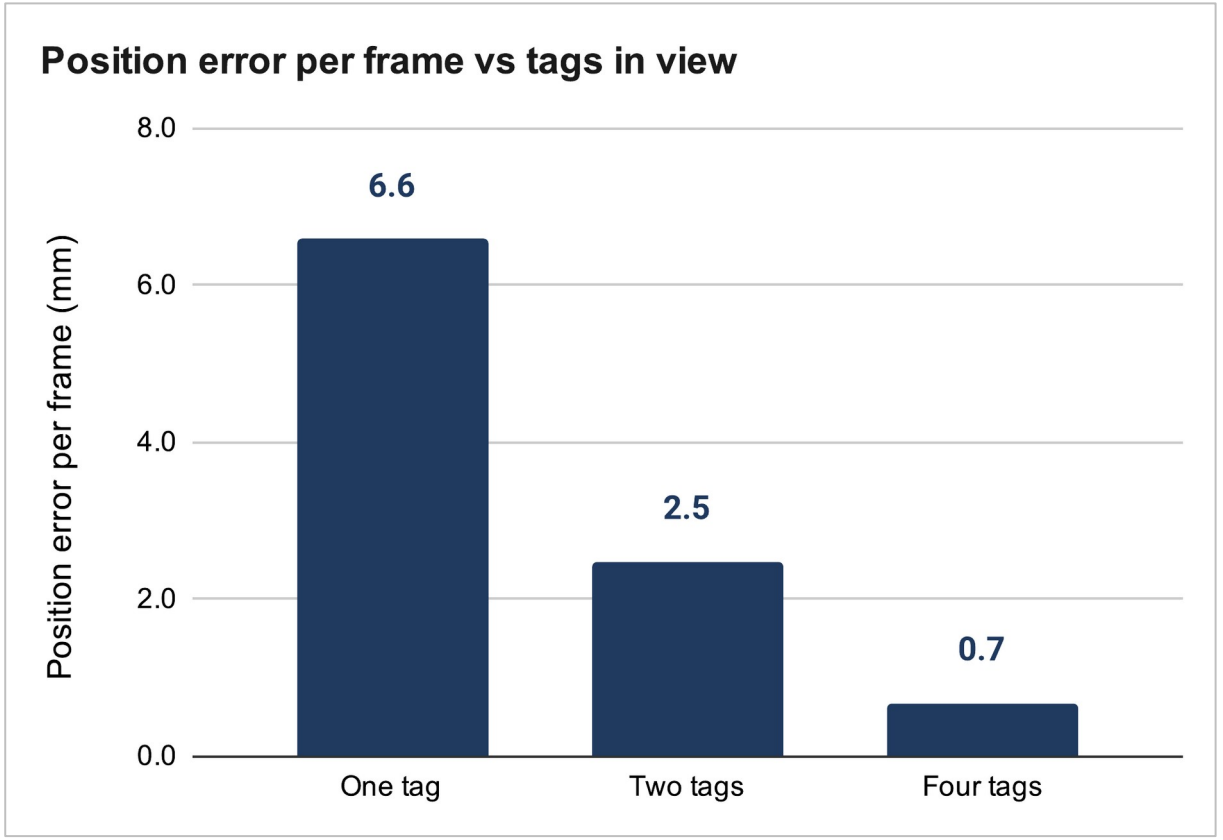
$$= \mathbf{2.0\ W} < 4.05\ \text{W}$$

where q_∞ = most heat any length can carry · w = 0.044 m · h ≈ 9.3 W/m²·K painted · k = 167 W/m·K (6061) · t = 0.000483 m · θ_b = 60 − 23 = 37 K

Floor Board Accuracy

Calculated

- Angle error × lever arm
- 1 tag: 6.6 mm · 4 tags: 0.7 mm
- Averaging removes noise, not bias



My calculation sheet (Google Sheets), “Floor board” tab. Calculated, not measured.

Equation

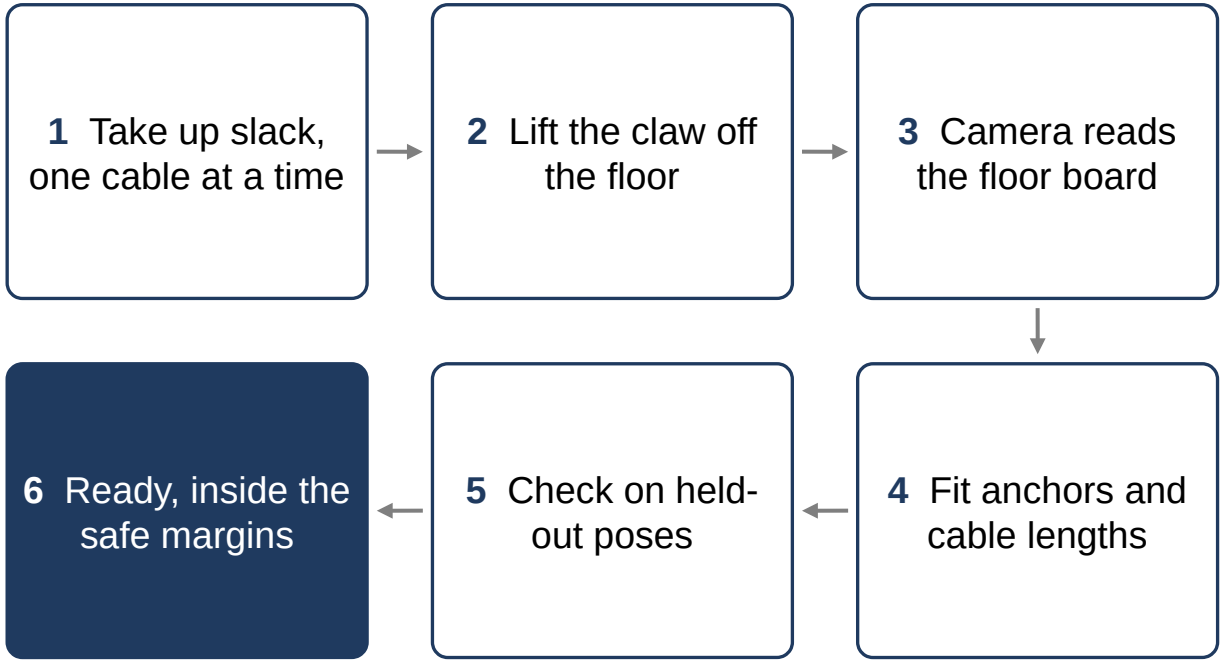
$$\delta x = \sigma_\theta \times L = 12.9\ \text{mrad} \times 0.51\ \text{m} = \mathbf{6.6\ mm}$$

where σ_θ = pose-angle noise with one tag in view · L = 0.51 m camera height above the board

Startup Calibration

Simulated

- No tape measure
- Fit: anchors + cable zero lengths
- Held-out poses check the fit
- Full room only inside error bars



Equation

$$\min \Sigma_k \Sigma_i \left(L_{i0} + d_{ik} - \|\mathbf{a}_i - \mathbf{p}_k\| \right)^2$$

$$\text{Full room only if } k \cdot \sigma \le \text{margin}$$

where a_i = anchor · p_k = claw pose k (camera) · d_{ik} = cable paid out (encoder) · L_{i0} = cable zero length · σ = fit error

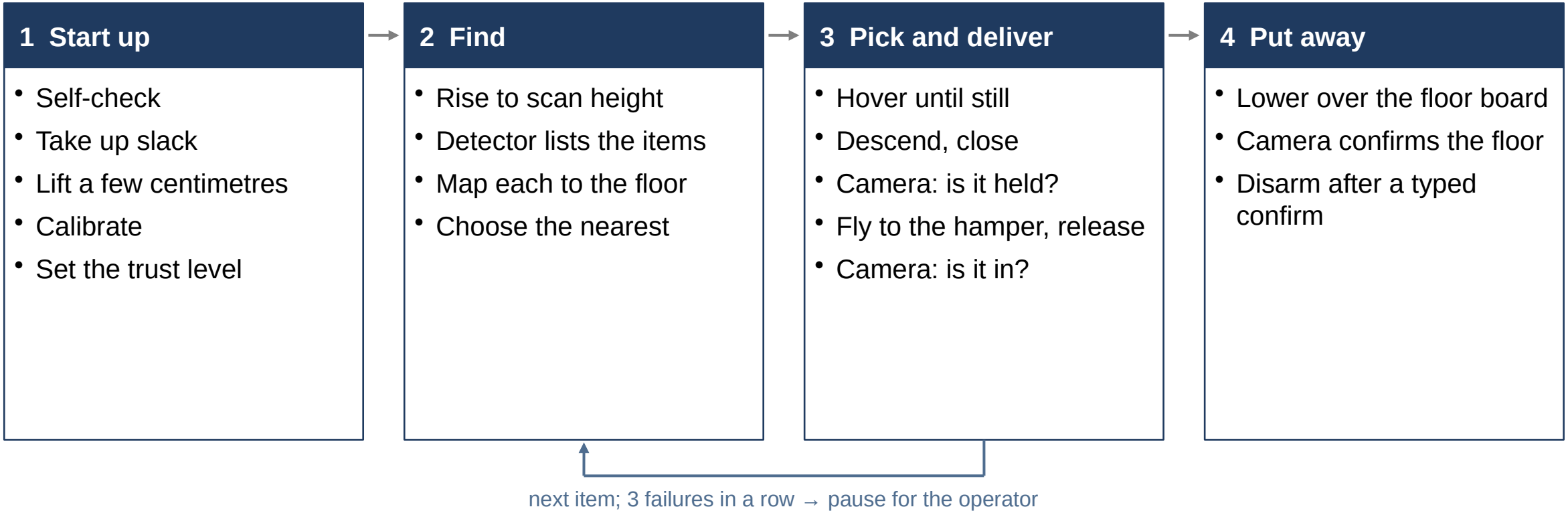
Logic: Gen 1 vs Final

Simulated

	Gen 1	Final
Where is the claw?	Assumed: where the cables were told to put it	Measured: claw camera on the floor board, refit at every startup
Room geometry	Point mass; anchors from a tape measure	Rigid claw; anchors and cable lengths fitted, with error bars
Who runs the motors	One central board steps all four	Each station runs time-stamped moves; the laptop coordinates
Finding laundry	Ceiling camera, pixel to floor map	Claw camera, ray from the claw's pose to the floor
Did the grab work?	Assumed once the claw closed	Camera checks pickup and delivery
Faults	Emergency stop only	Supervisor: a stop keeps torque; nothing restarts by itself

One Run

Simulated

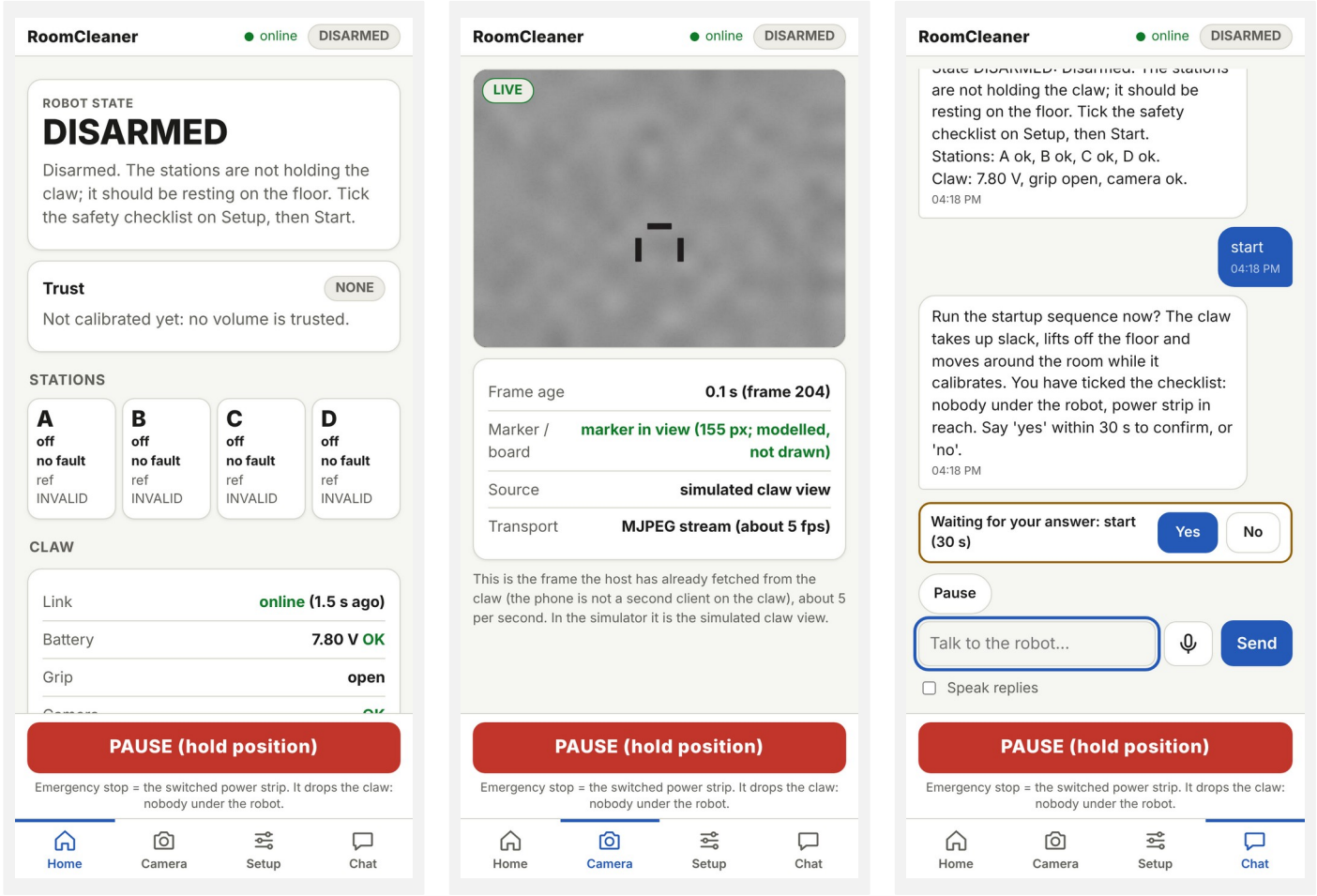


At any point: PAUSE or a fault stops every station on the same point of the path, still holding. Nothing restarts by itself.

Phone App

Simulated

- Status, live claw camera, guided setup
- Anything that moves asks “yes” first
- PAUSE holds position at once
- The switched power strip is the hard stop
- A reboot never resumes a task

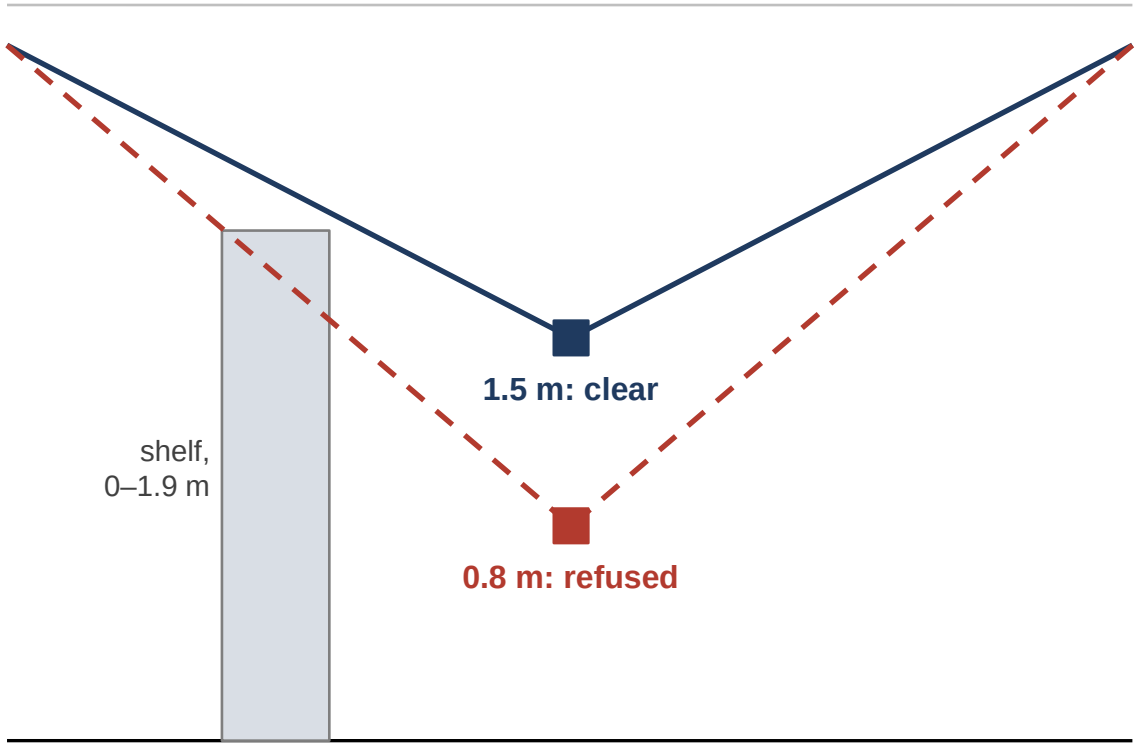


Phone app: home, camera, chat

Obstacles

Simulated

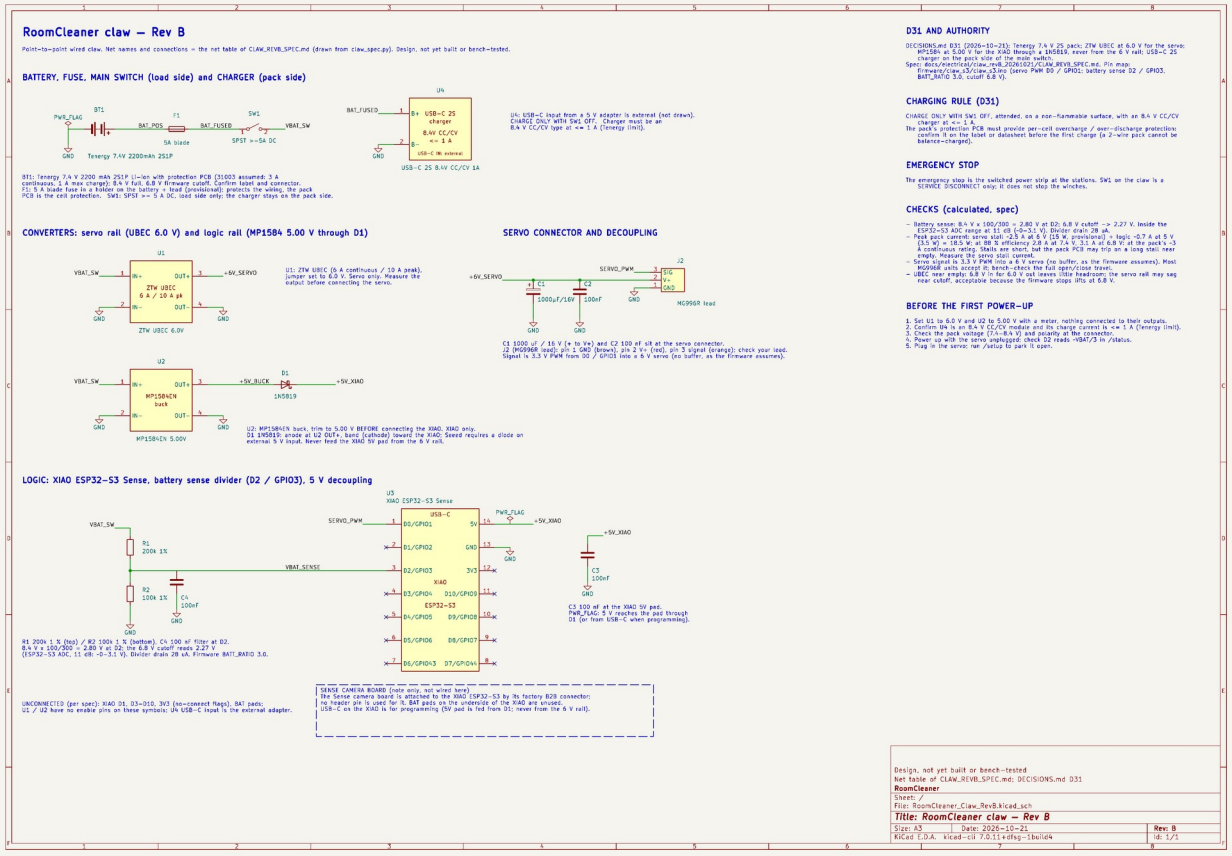
- Known obstacles = keep-out cylinders
- Every cable checked at every path point
- Fails → refused before moving
- Can't see people yet: nobody underneath



Side view along a room diagonal: same floor spot, two heights; an example shelf

Claw Electrical

- 2-cell lithium-ion pack: 7.4 V, 5 A fuse
- 6 V servo rail · separate 5 V logic
- Peak 2.8–3.1 A vs 3 A pack
- Charge: switch off, attended, ≤ 1 A



Gear Strength

Calculated

- Module 1.5 · PETG allowable 22.5 MPa
- 6 V stall: SF 2.05 / 2.50 / 2.00
- Module 1: SF 0.35–0.41 → rejected





Equation

$$F_t = 2\tau / d = 2 \times 1.078 / 0.027 = 79.9\ \text{N}$$

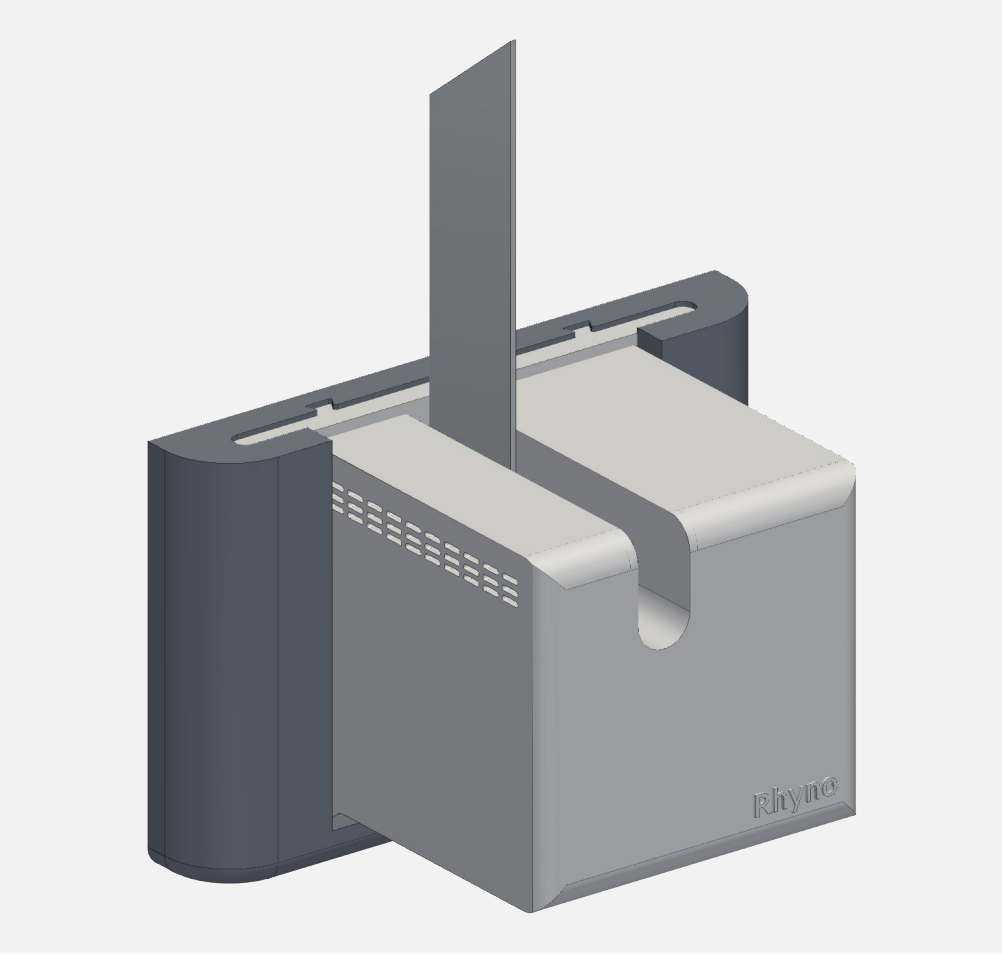
$$\sigma = F_t / (b\ m\ Y) = 79.9 / (15.7 \times 1.5 \times 0.309) = 11.0\ \text{MPa}$$

$$SF = \sigma_{allow} / \sigma = 22.5 / 11.0 = \mathbf{2.05}$$

where SF = safety factor · τ = servo stall torque · d = pitch diameter (18 teeth × 1.5 mm) · b = face width · m = module · Y = Lewis form factor

Safety

- One hard stop: switched power strip
- Faults and PAUSE hold torque
- Anchors in studs · nobody underneath
- Never unplug a motor under power

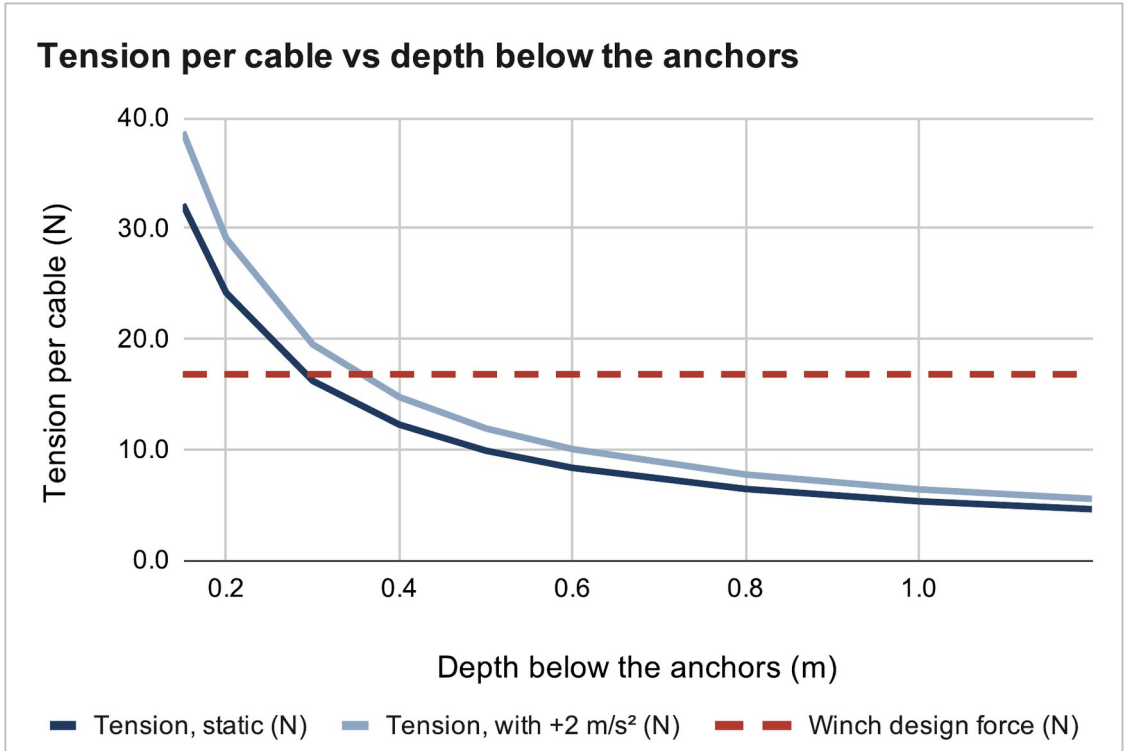


Acronyms

Term	Meaning
CAD	Computer-aided design
PETG	Polyethylene terephthalate glycol (the printed plastic)
TPU	Thermoplastic polyurethane (flexible printed plastic)
NEMA 17	National Electrical Manufacturers Association motor frame, 1.7 in square
I²C	Inter-Integrated Circuit: a short-range two-wire sensor bus
ESP32	A Wi-Fi microcontroller family (Espressif)
A4988 · AS5600	Stepper-motor driver chip (Allegro) · magnetic angle sensor (ams)
YOLO	“You Only Look Once”: an object-detection model family
SF	Safety factor: allowable stress ÷ actual stress

Calculations and Sources

- Every chart: my calculation sheet
- Every result: a formula on marked inputs
- **Sources:**
  - Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3
  - Allegro MicroSystems, A4988 datasheet



My calculation sheet (Google Sheets), “Cable tension” tab. Calculated, not measured.

RoomCleaner engineering calculations (Google Sheets)